

# FI 362 — ADVANCED FINANCIAL ECONOMETRICS

Mock Midterm Examination |

Financial Time Series & Linear Time Series Analysis

Practice paper

**Time allowed:** 120 minutes

**Total:** 100 points

**Part A:** 5 multiple-choice ( $5 \times 10 = 50$  pts) **Part B:** 1 solution/interpretation question (50 pts)

*Instructions.* Answer *all* questions. For Part A circle exactly one letter per question. For Part B show your reasoning: numerical answers with no interpretation earn little credit. Calculators permitted; no notes. Useful values:  $1.96/\sqrt{T}$  for the ACF band;  $\chi^2_{2,0.05} = 5.99$ ,  $\chi^2_{10,0.05} = 18.31$ ; Dickey–Fuller 5% critical values  $-2.86$  (with drift) and  $-1.94$  (no drift).

*This is a practice paper. It mirrors the format, coverage and difficulty of the graded midterm (Chapters 1–2 only); the numbers and scenarios are new.*

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## Part A — Multiple Choice (10 points each)

- A1.** Let  $R_t$  be the simple net return and  $r_t = \ln(1 + R_t)$  the log return of an asset, and let a portfolio hold weights  $w_i$  in assets  $i$ . Which statement is **true**?
- (A) Log returns aggregate *exactly* across the assets of a portfolio,  $r_{p,t} = \sum_i w_i r_{i,t}$ .
  - (B) Simple returns aggregate *additively* over time,  $R_t(k) = R_t + R_{t-1} + \cdots + R_{t-k+1}$ .
  - (C) Simple returns aggregate *exactly* across the assets of a portfolio,  $R_{p,t} = \sum_i w_i R_{i,t}$ .
  - (D) For any given price move the log return is always *larger* than the simple return.
- A2.** A daily return series has sample skewness  $\approx -0.6$  and sample *excess* kurtosis  $\approx +6.5$ . The most accurate reading is:
- (A) The distribution has *fat tails* and a *heavier left tail*: extreme moves — especially large *losses* — are more likely than a Normal with the same variance would predict.
  - (B) The returns are thinner-tailed than the Normal, so large moves are rare.
  - (C) The positive number (+6.5) is a skewness measure implying frequent large *gains*.
  - (D) Excess kurtosis of +6.5 equals the Normal benchmark, so normality holds.
- A3.** A stationary series has a sample ACF that *decays geometrically* (tails off), while its PACF is significant at lag 1 and negligible from lag 2 onward (cuts off). The most appropriate model is:
- (A) MA(1).
  - (B) ARMA(1,1).
  - (C) AR(2).
  - (D) AR(1).

- A4.** Consider the model  $x_t = 0.5 + 0.9x_{t-1} + a_t$  with  $a_t \sim \text{WN}(0, \sigma_a^2)$ . Which statement is **true**?
- (A) It is non-stationary because the AR coefficient 0.9 is close to 1.
  - (B) It is weakly stationary with unconditional mean  $\mathbb{E}[x_t] = 0.5/(1 - 0.9) = 5$ , and its ACF decays geometrically as  $\rho_k = 0.9^k$ .
  - (C) Its long-run mean equals the intercept,  $\mathbb{E}[x_t] = 0.5$ .
  - (D) Its ACF cuts off to zero after lag 1.
- A5.** Two *independent*  $I(1)$  series are regressed on each other by OLS. The output shows  $R^2 = 0.92$  and a slope  $t$ -statistic of 210, but the residuals have a lag-1 autocorrelation of 0.98. This is best described as:
- (A) A genuine cointegrating relationship, confirmed by the high  $R^2$ .
  - (B) Proof that the two series are white noise.
  - (C) Evidence that the OLS standard errors are *too large*, understating the true significance.
  - (D) A spurious regression: the OLS  $t$ -statistic is unreliable because the near-unit-root residuals make its standard error far too *small*; difference the data or use HAC (Newey–West) standard errors.

## Part B — Solution & Interpretation (50 points)

An analyst studies  $T = 1600$  daily log returns (in decimal) of an equity index ETF ABC. The computer output is summarised below.

**Table 1 — Descriptive statistics of the daily log returns**

| mean   | sd    | skewness | excess kurtosis | Jarque–Bera    | JB $p$ -value |
|--------|-------|----------|-----------------|----------------|---------------|
| 0.0003 | 0.011 | −0.60    | 6.5             | $\approx 2913$ | $< 0.001$     |

**Table 2 — Autocorrelation diagnostics**

|                         | lag-1 autocorrelation              | Ljung–Box $Q(10)$ ( $p$ -value) |
|-------------------------|------------------------------------|---------------------------------|
| returns $r_t$           | +0.062                             | 31.5 ( $< 0.001$ )              |
| squared returns $r_t^2$ | +0.30 (and still $> 0$ at lag 150) | 2600 ( $< 0.001$ )              |

**Table 3 — Augmented Dickey–Fuller tests**

| series                               | ADF $t$ -statistic | 5% critical value |
|--------------------------------------|--------------------|-------------------|
| log <i>price</i> of ABC (with drift) | −1.85              | −2.86             |
| log <i>return</i> of ABC (no drift)  | −28.7              | −1.94             |

- (a) **(12 pts)** From Table 1 the Jarque–Bera statistic is  $JB \approx 2913$ .
- (i) Compare it with the critical value  $\chi_{2,0.05}^2 = 5.99$ : are the returns plausibly Normal? Reject or not?

- (ii) The excess kurtosis is +6.5. In one sentence, what does a large *positive* excess kurtosis say about the tails, and which stylised fact of returns is this?
  - (iii) The skewness is  $-0.60$ . What does the *negative* sign say about up-moves versus down-moves?
- (b) **(12 pts)** With  $T = 1600$  the ACF significance band is  $\pm 1.96/\sqrt{1600} = \pm 0.049$ .
- (i) Is the return's lag-1 autocorrelation of  $+0.062$  inside or outside the band — i.e. is it statistically significant?
  - (ii) The one-step  $R^2$  of an AR(1) equals  $\rho_1^2$ . Compute it as a percentage. Is the predictable part of daily returns economically large or tiny?
  - (iii) Name one market feature (microstructure) that can create a small lag-1 autocorrelation in an *index* return series.
- (c) **(12 pts)** Read Table 3.
- (i) For the log *price*, do we reject the unit-root null? Is the price  $I(0)$  or  $I(1)$ ?
  - (ii) For the log *return*, do we reject? Is it  $I(0)$  or  $I(1)$ ?
  - (iii) In one sentence, why do we model *returns* rather than prices?
- (d) **(14 pts)** Compare the two rows of Table 2.
- (i) Which ACF dies out almost at once, and which stays large for hundreds of lags?
  - (ii) Is it the *direction* (mean) or the *size/volatility* (variance) of returns that is predictable? One sentence.
  - (iii) Which class of model is designed to capture the pattern in the *squared* returns (name the family)?

*Hint / formulas.* Reject Normality if  $JB > \chi_{2,0.05}^2 = 5.99$ . A correlation is significant if it falls *outside* the band  $\pm 1.96/\sqrt{T}$ . ADF: reject the unit root (conclude *stationary*) if the  $t$ -statistic is *below* (more negative than) the Dickey–Fuller critical value. The Normal distribution has excess kurtosis 0; positive excess kurtosis = fatter tails.

## SOLUTIONS & MARKING GUIDE

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### Part A — Answer key

| Question | A1 | A2 | A3 | A4 | A5 |
|----------|----|----|----|----|----|
| Answer   | C  | A  | D  | B  | D  |

- A1. (C)** *Simple* returns aggregate exactly *across assets*:  $R_{p,t} = \sum_i w_i R_{i,t}$ , because portfolio wealth is a weighted sum of the constituent gross returns. (A) is false — log returns do this only *approximately*, since the log of a weighted sum is not the weighted sum of logs. (B) is false: simple returns compound *multiplicatively* over time,  $1 + R_t(k) = \prod_{j=0}^{k-1} (1 + R_{t-j})$ ; it is *log* returns that are time-additive. (D) is false: because  $\ln(1+x) \leq x$ , the log return is always  $\leq$  the simple return.
- A2. (A)** The Normal has excess kurtosis 0; a large *positive* value (+6.5) means *fatter tails* than the Normal, so extreme moves are more frequent, and the *negative* skew (−0.6) tilts that extra tail mass toward large *losses*. (B) reverses the tail logic; (C) confuses kurtosis with skewness *and* the sign; (D) confuses excess kurtosis (Normal = 0) with raw kurtosis (Normal = 3).
- A3. (D)** An ACF that *tails off* (geometric decay) with a PACF that *cuts off* after lag 1 is the textbook signature of an **AR(1)**. MA(1) shows the mirror pattern (ACF cuts off at lag 1, PACF tails off); ARMA(1,1) has *both* tailing off; AR(2) would have the PACF cut off after lag 2.
- A4. (B)** With  $|\phi_1| = 0.9 < 1$  the AR(1) is weakly stationary. Its unconditional mean is  $\mathbb{E}[x_t] = \phi_0 / (1 - \phi_1) = 0.5 / 0.1 = 5$  (*not* the intercept 0.5, ruling out (C)), and its ACF decays geometrically  $\rho_k = 0.9^k$  (ruling out (D), which describes an MA(1)). (A) confuses “close to 1” with “equal to 1”: only  $\phi_1 = 1$  (a unit root) is non-stationary.
- A5. (D)** Regressing two independent  $I(1)$  series gives a **spurious regression**: the high  $R^2/t$  are artefacts of the strongly autocorrelated ( $\hat{\rho}_1 = 0.98$ ) residuals, which make the OLS standard error far *too small* (so (C) has the bias backwards). (A) is wrong — independent series are not cointegrated, and their residuals are non-stationary; (B) is nonsense. Remedy: difference both series (regress the returns) or report HAC / Newey–West standard errors.

### Part B — Worked solution

**(a) Distribution / normality (12 pts).** **(i)**  $JB \approx 2913$  is vastly bigger than  $\chi_{2,0.05}^2 = 5.99$ , so we **reject normality** (consistent with  $p < 0.001$ ). **(ii)** The Normal has excess kurtosis 0; a large *positive* value (+6.5) means **fat tails** — extreme moves (crashes, jumps) are far more common than a Normal allows (the heavy-tails stylised fact). **(iii)** The *negative* skew (−0.6) means large *down*-moves are sharper / more frequent than comparable up-moves (mild “leverage” asymmetry). (*Verification: with  $T/6 = 266.67$ , the skewness term is  $266.67 \times 0.6^2 = 96$  and the kurtosis term is  $266.67 \times 6.5^2 / 4 = 2817$ , giving  $JB \approx 96 + 2817 = 2913$ ; the kurtosis term dominates.*)

**(b) Predictability of the mean (12 pts).** (i) Band =  $\pm 1.96/\sqrt{1600} = \pm 0.049$ ; since  $0.062 > 0.049$ , the lag-1 autocorrelation is **outside the band, i.e. statistically significant** (and  $Q(10) = 31.5 > 18.31$  agrees). (ii)  $R^2 = \rho_1^2 = (0.062)^2 = 0.003844 \approx \mathbf{0.38\%}$  — *tiny*: yesterday explains only about a third of one percent of today's return, no usable edge after costs, so returns are economically almost unpredictable even though the correlation is statistically real. (iii) In an *index*, small lag-1 autocorrelation typically comes from **non-synchronous (stale-price) trading** — constituent stocks trade at slightly different times, so the index price partially reflects yesterday's information, inducing autocorrelation. (The bid-ask bounce is the analogous single-stock microstructure effect and earns full credit.)

**(c) Order of integration (12 pts).** (i) Log *price*:  $-1.85 > -2.86$ , so we **fail to reject** the unit root — the price is **I(1)**. (ii) Log *return*:  $-28.7 < -1.94$ , so we **reject** — the return is **I(0)**, stationary. (iii) We model returns because they are stationary ( $I(0)$ ), the prerequisite for valid estimation and inference, whereas the price wanders (unit root) and cannot be modelled directly — one difference of the log price *is* the log return.

**(d) Predictability of the variance (14 pts).** (i) The ACF of the *returns*  $r_t$  dies out almost immediately; the ACF of the *squared* returns  $r_t^2$  stays large for hundreds of lags. (ii) So it is the **size/volatility (variance), not the direction (mean), that is predictable** — big moves cluster with big moves, calm with calm (volatility clustering). (iii) Linear AR/MA/ARMA act on the mean and cannot capture this; the **ARCH/GARCH** family (Chapter 3) is built for persistent, forecastable volatility.